CNENGR202 Instrumentation and Control

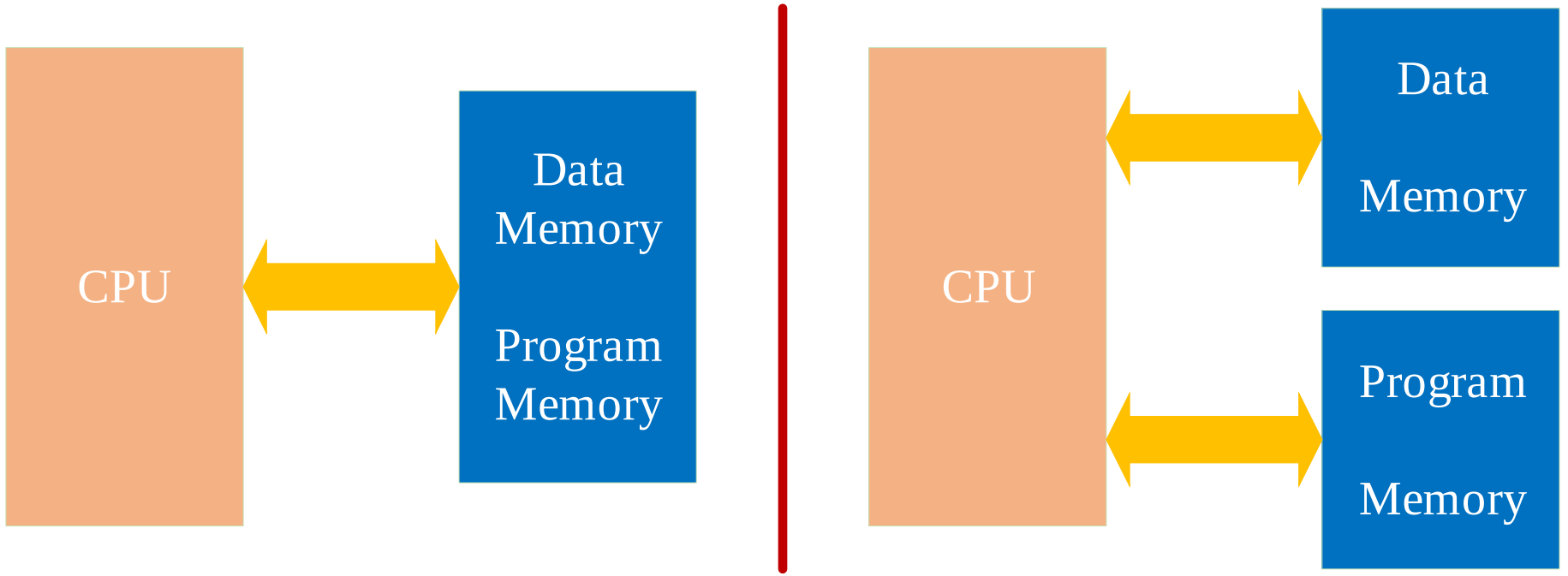
Part 3 L3 – Memory

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Recap



Von Neumann

Harvard

Outline

- So far we've written numbers to registers.
- Not very useful

- Perform **useful** operations
  - Mathematical (add, sub etc)
  - Logical (and, or, xor…)

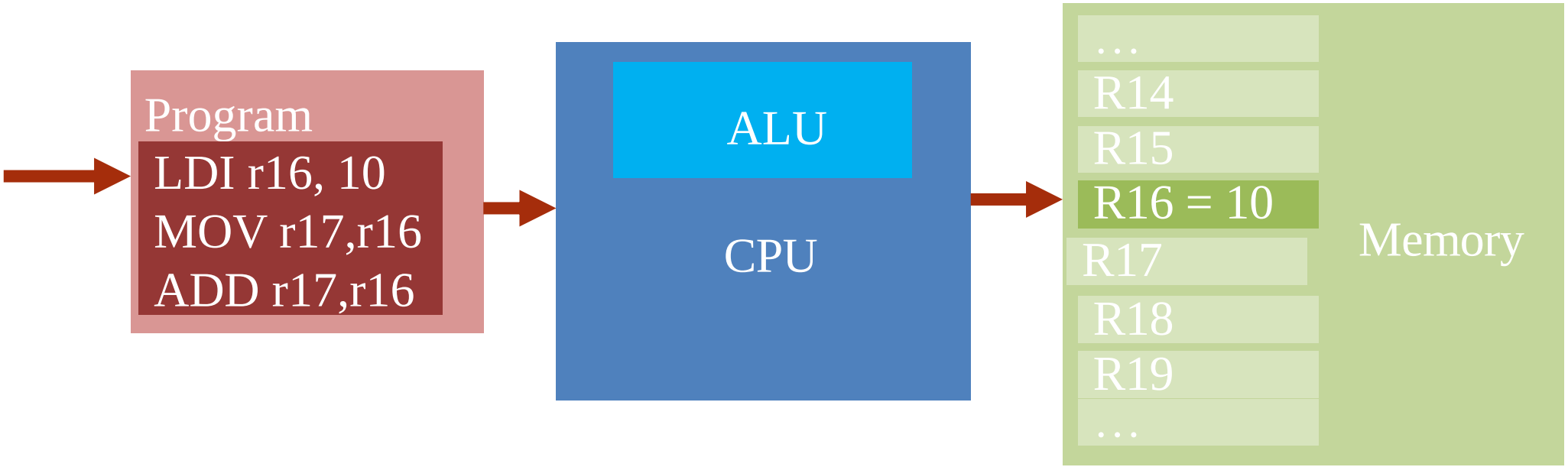
- Input/Output
- SRAM

ALU (Arithmetic Logic Unit)

- Takes care of all mathematical and logical operations
- All simple mathematical operations commands take similar form
- **ADD Rd,Rr** will add **Rr** to **Rd** and leave the answer in **Rd**.
  - Rd = Rd + Rr
  - *ADD r17, r16*

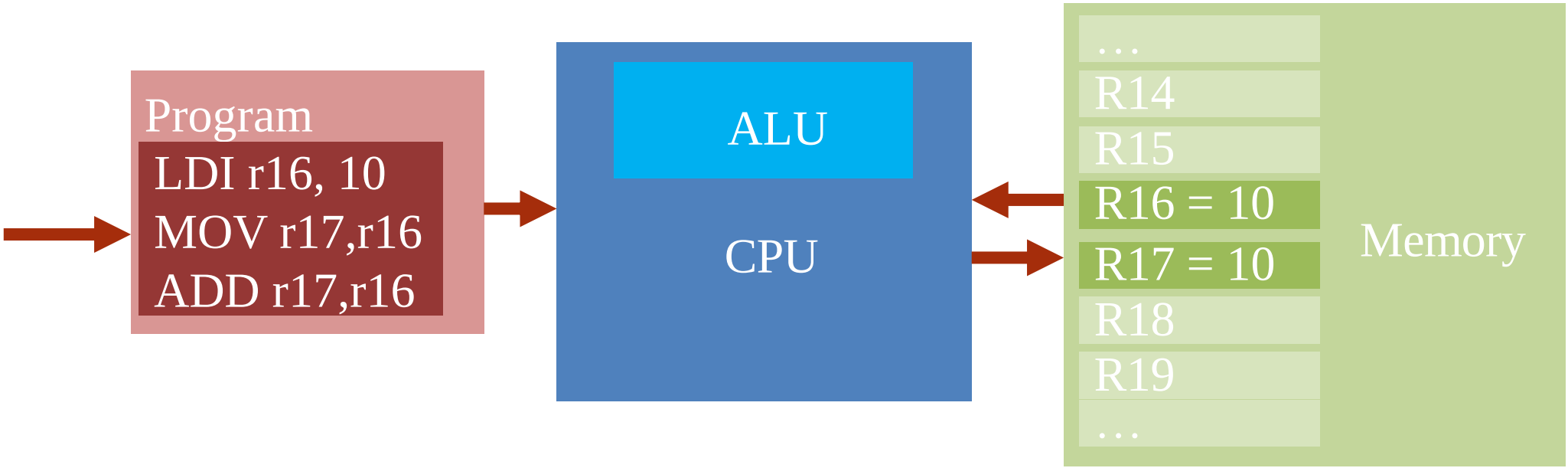
Adding

- Set a value to a register (10 to r16)



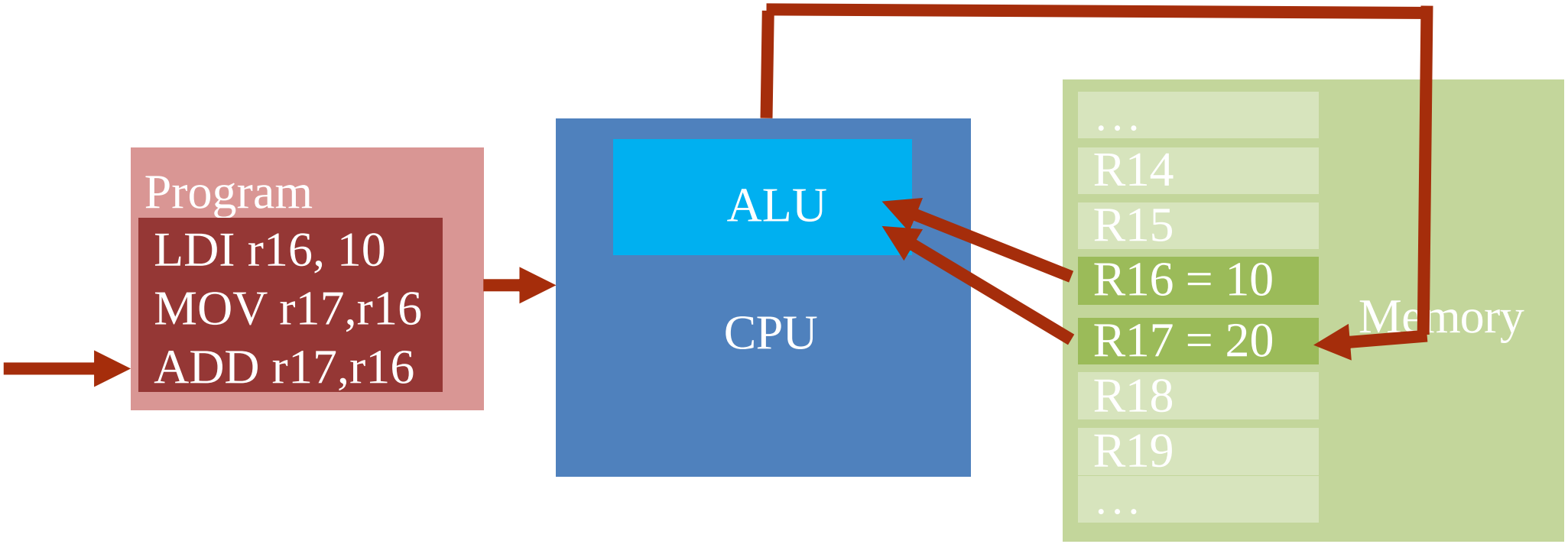
Adding

- Copy this to register 17
- We have two registers with values in them…

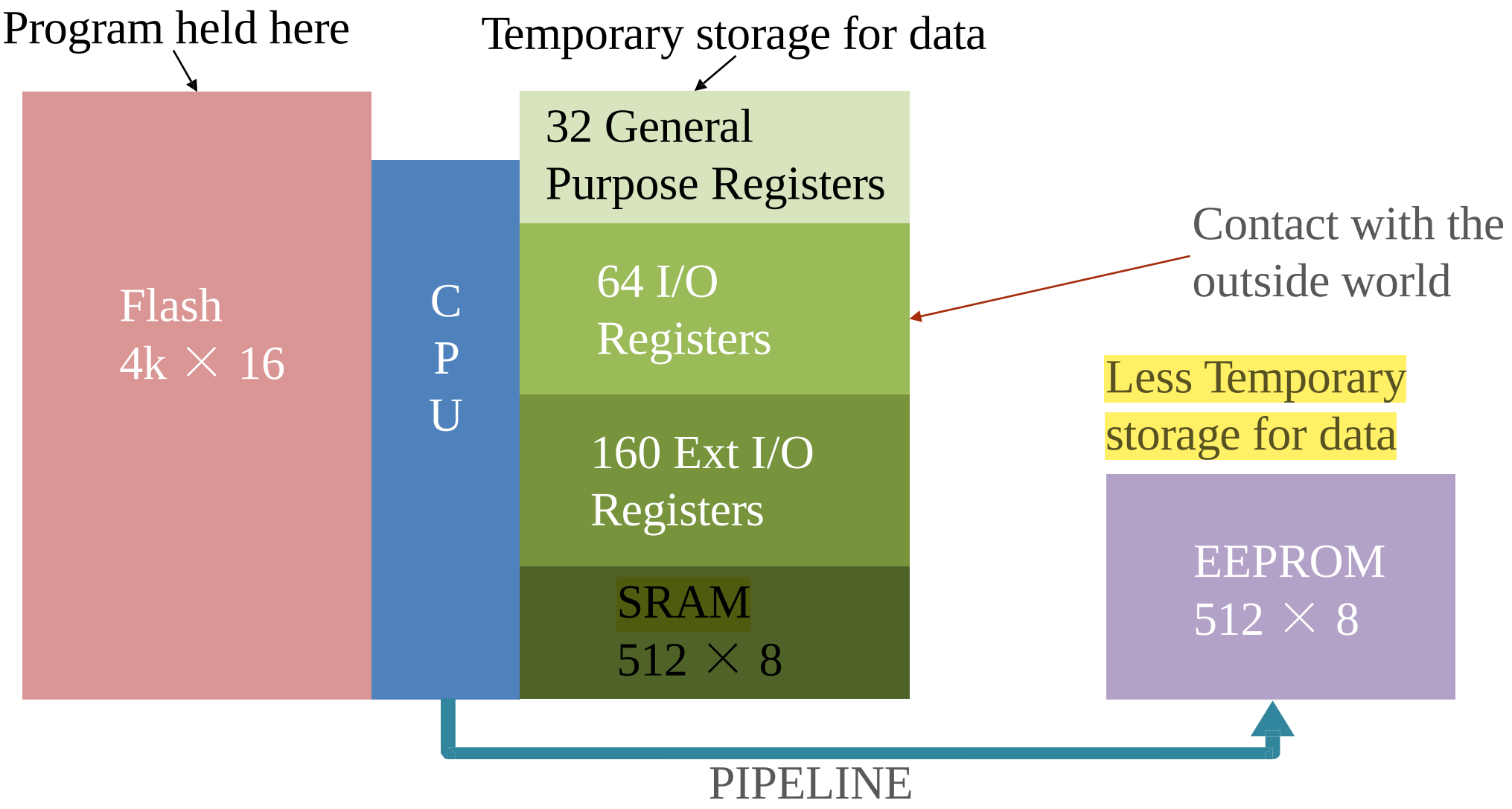


Adding

- Perform the add leaving the result in r17



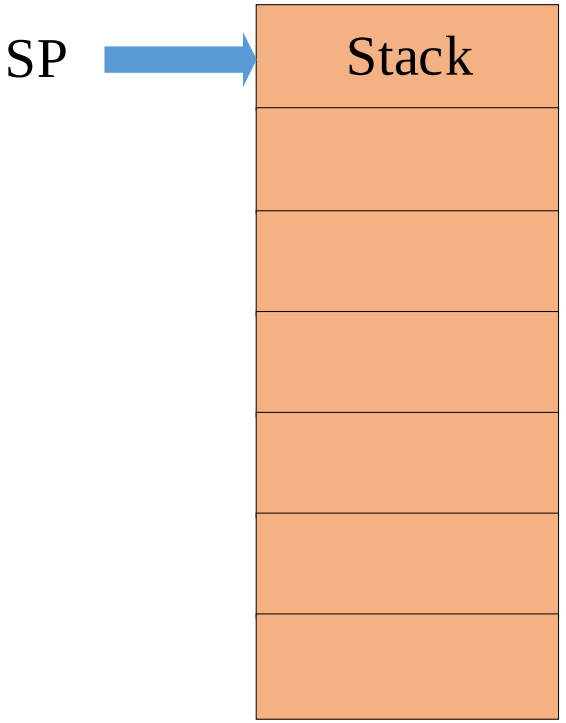
ATMega Memory - map



Stack - I

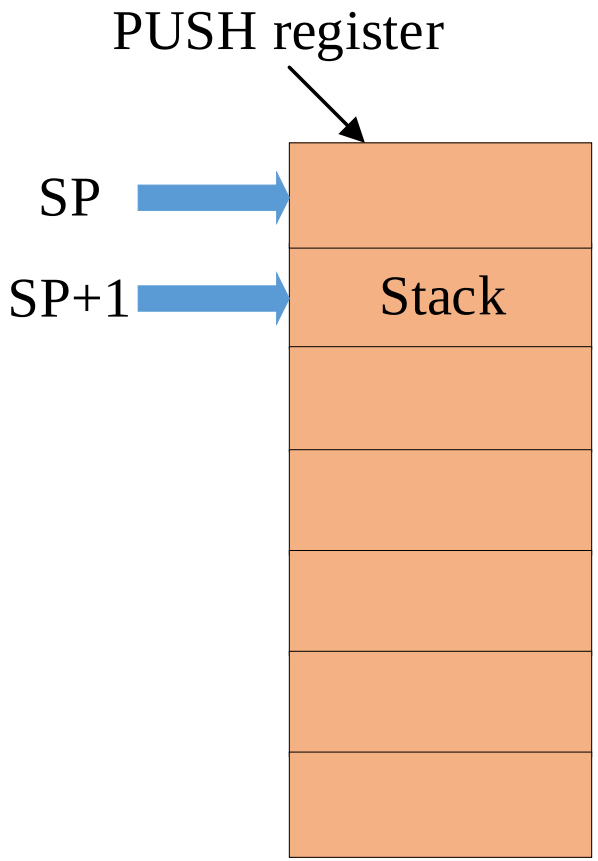


- The stack lives in the SRAM
  - 512 bytes
- Last - In - First - Out (LIFO)
- List of 8bit chunks of data
- This can be a good place to pop things for a moment.



Stack - II

- Top of the stack is tracked by the stack pointer (SP)



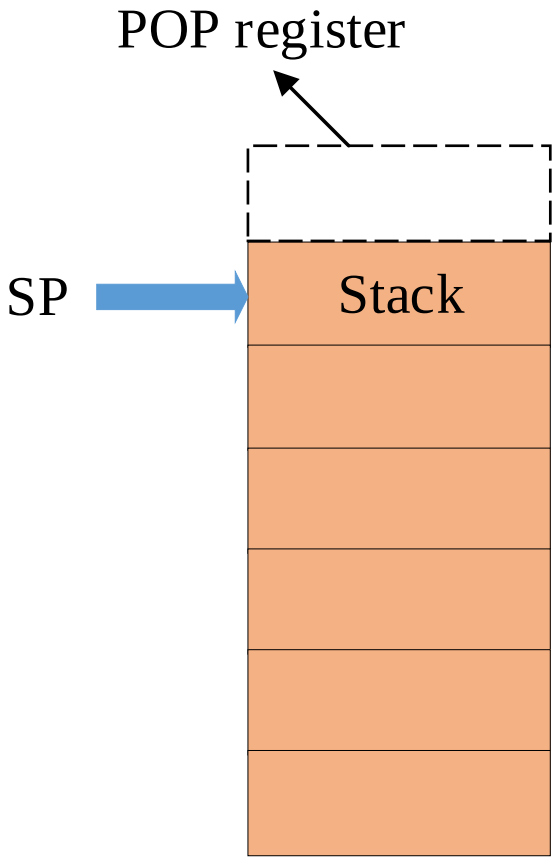
- It is manipulated by two commands
  - PUSH Rr
- Pushes the value of a register onto the top of the stack
- Increments the stack pointer

Stack - III

- Top of the stack is tracked by the stack pointer (SP)

- It is manipulated by two commands
  - POP Rd
- POPs the value off the top f the stack into a register

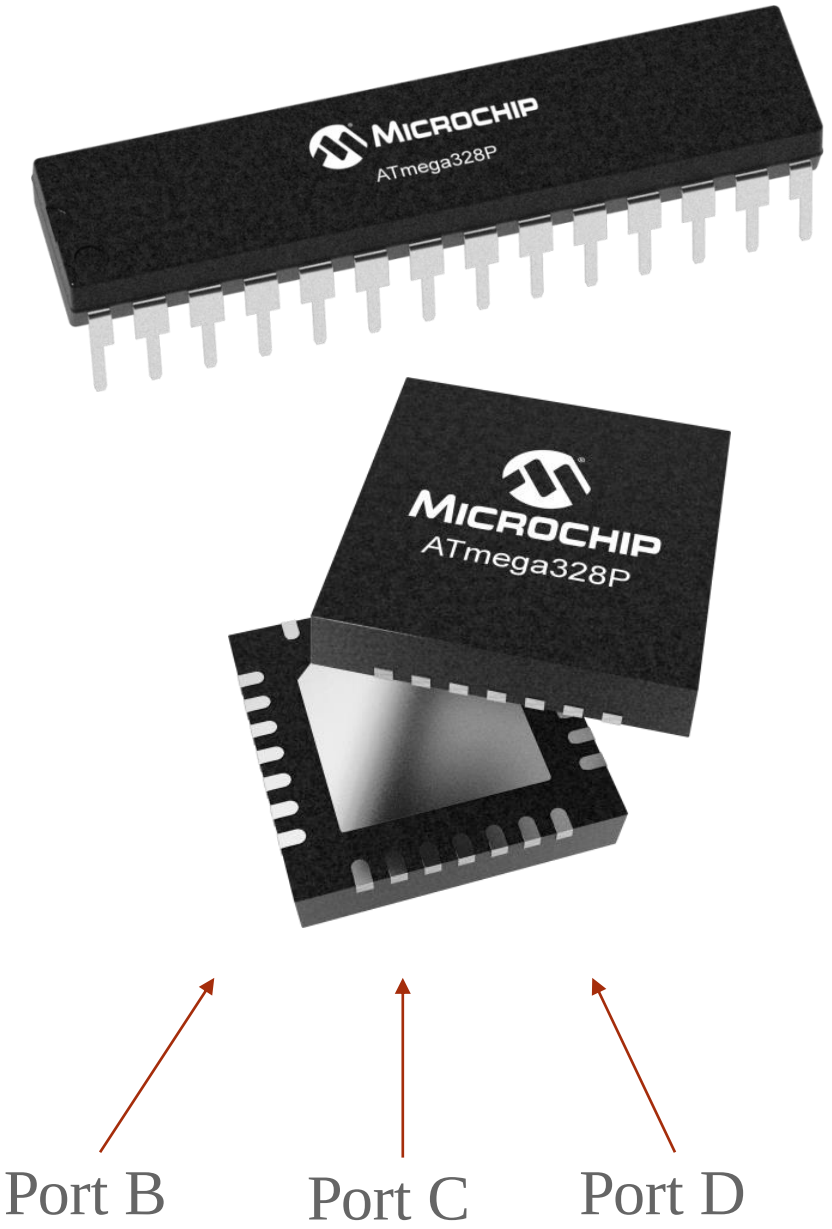
- Decrements the stack pointer



Ports - I

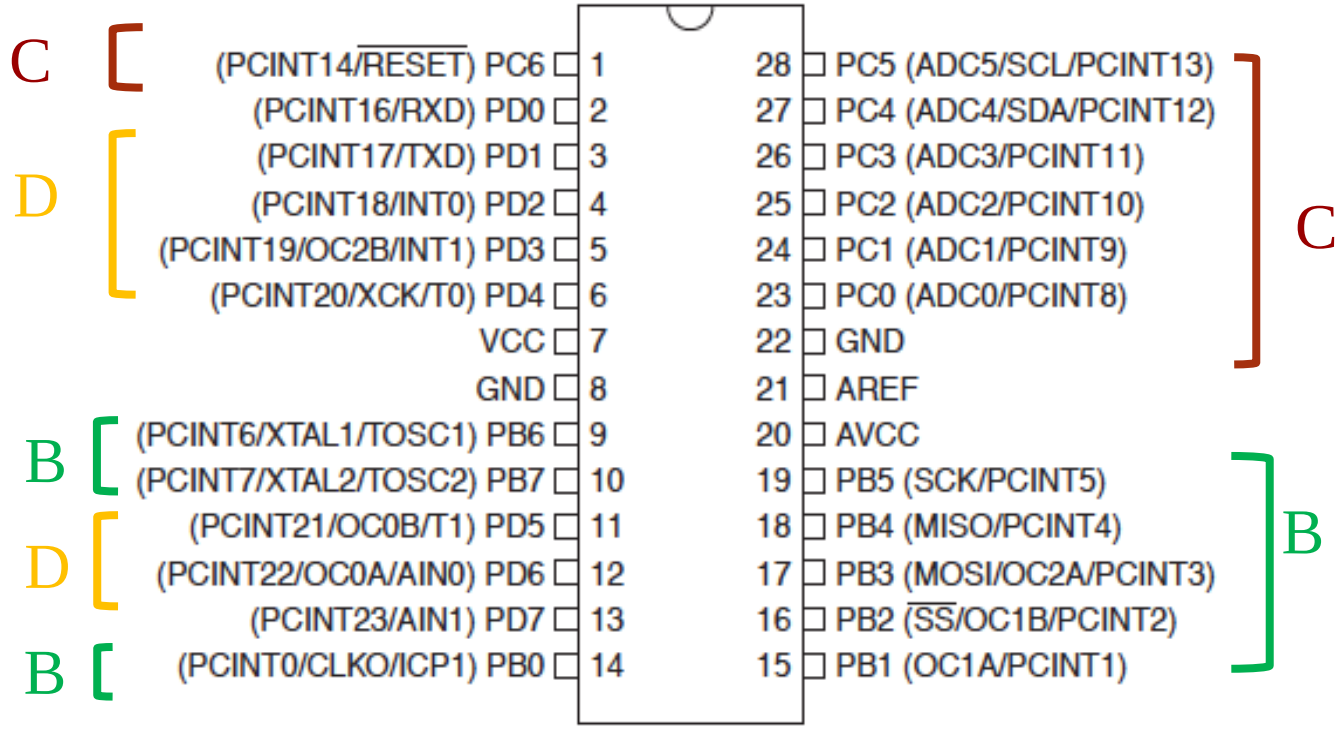
A port is a collection of 8 **physical** pins

- Ports are the windows to communicate with the world
- Allow the flow of data into and out of the chip to the surrounding circuit
- Bi - directional (in and out)



Ports - II

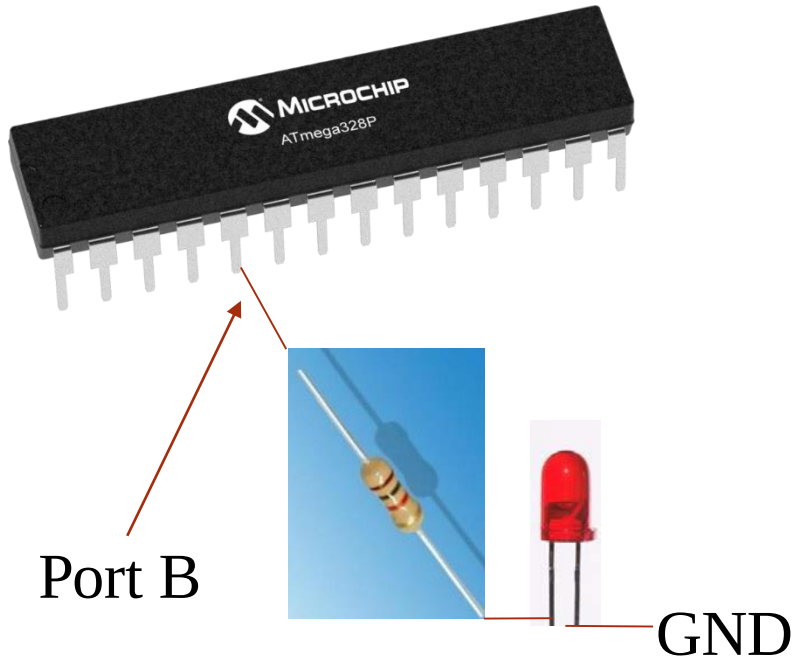
- The ATMega 328P using has three ports, B - D.
- Physically pins are split up
- No port A…



Ports - III

- Each port has a 8 bit register tied to it
  - For PortB this register is called PortB
    - Looks just like a general purpose register
  - This register can **control** the pin voltage
    - Use as an output to turn on LEDs
  - Or it can be **controlled by** the pin voltage
    - Read in voltage signals (from a switch or…)

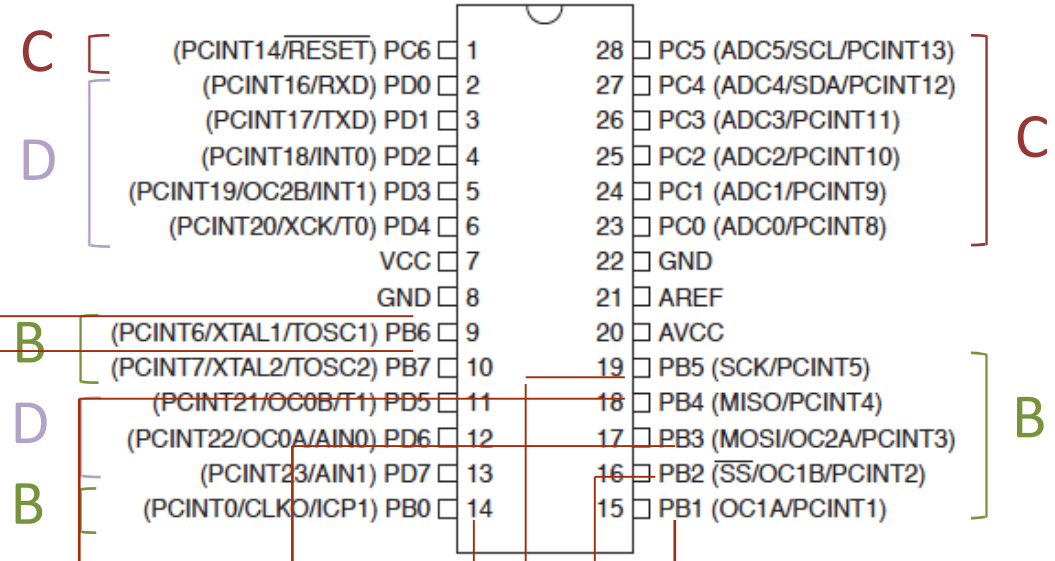
Physically



PORTB- The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05(0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Register to pin connections



PORTB- The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05(0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Data Direction Register

- Every port has an 8 bit data direction register or DDR.
- For Port**B** this is DDR**B**
- Every bit corresponds to a pin on the port and designates it as:
  - 1 = output
  - 0 = input

DDRB- The Port B Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x04(0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB

- Inputs and outputs can be mixed on a port
- If you **read an output** pin or **write to an input** pin you might get an unexpected result

Port Commands

- When we move to a different area of the memory map you need new commands.
  - *LDI, MOV, ADD* etc. no longer work
- Values cannot be written directly to ports
  - *LDI PortB, 34* // this is not valid

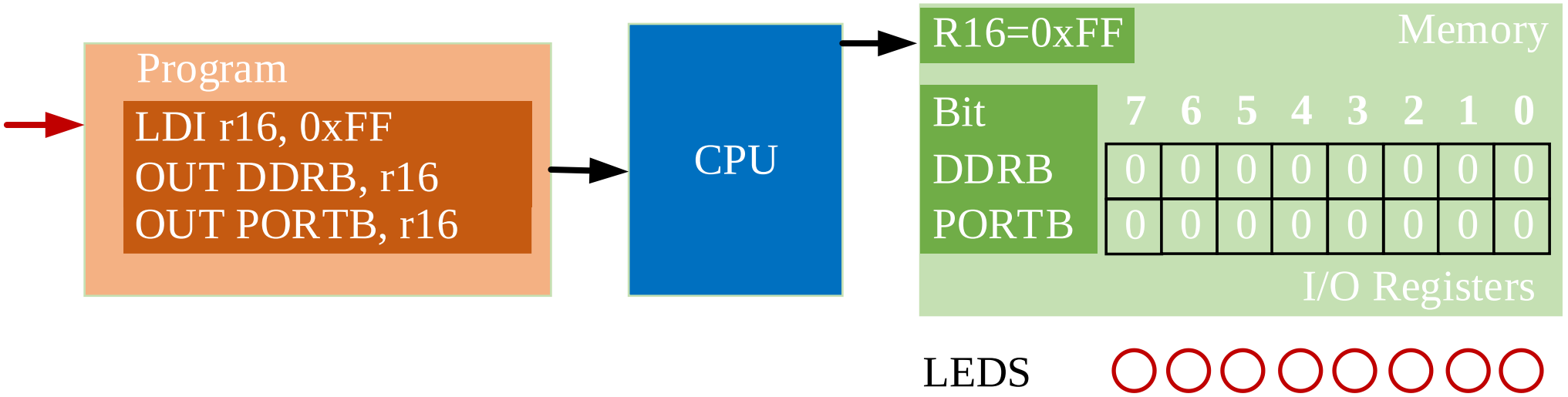
- Use general purpose registers for program logic
- Then **copy** to port

Write to a port - I

- To write to a port use OUT
- Write value from r16 to PortB
  - *OUT PortB, r16*

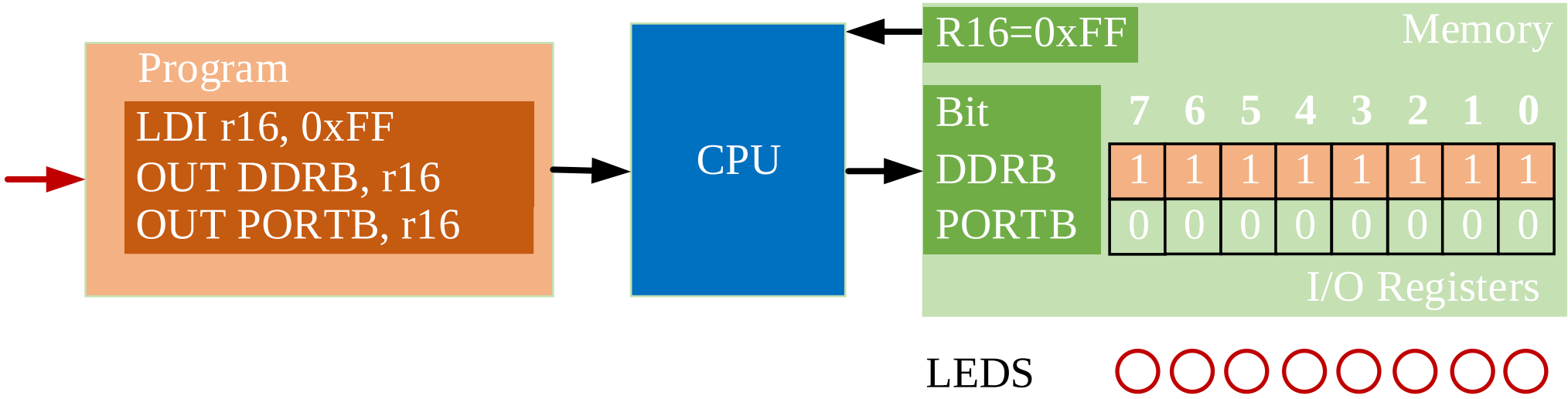
Write to a port - II

- Write 0b11111111 or 0xFF or 255 to r16



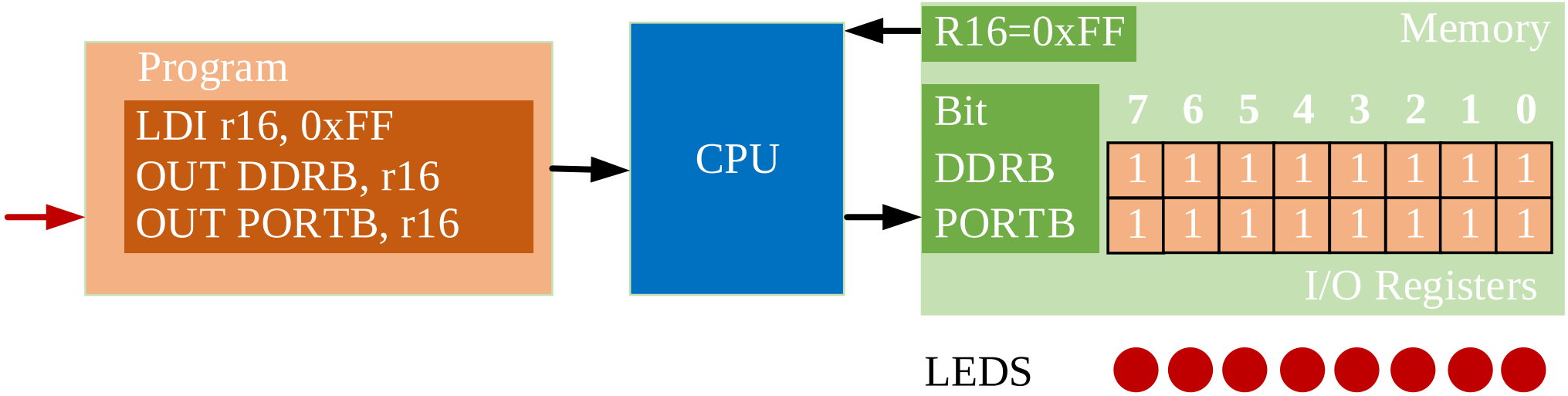
Write to a port - III

- Set all pins to outputs



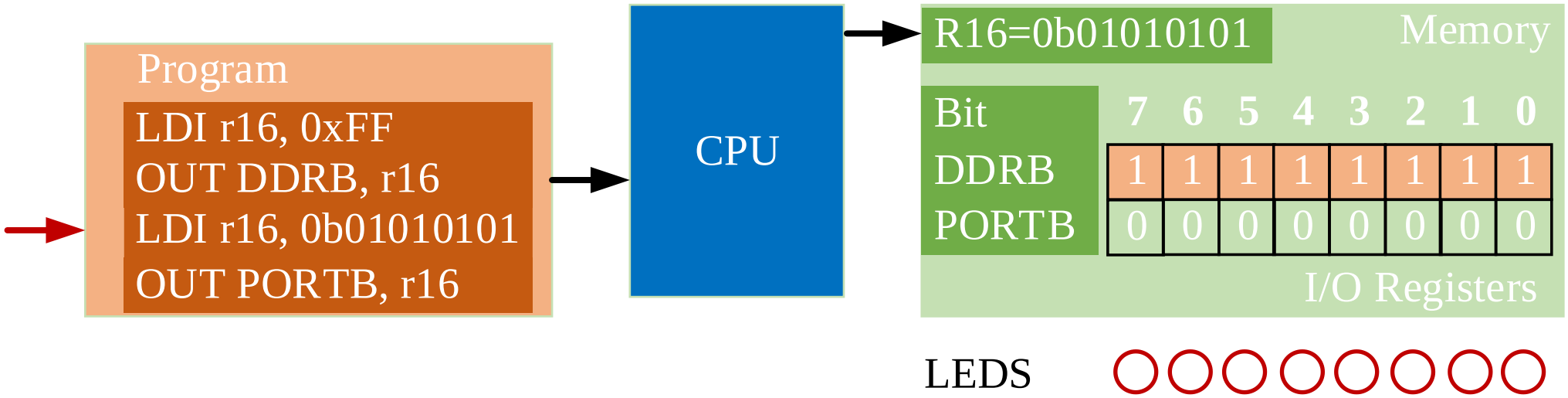
Write to a port - IV

- Turn all pins on



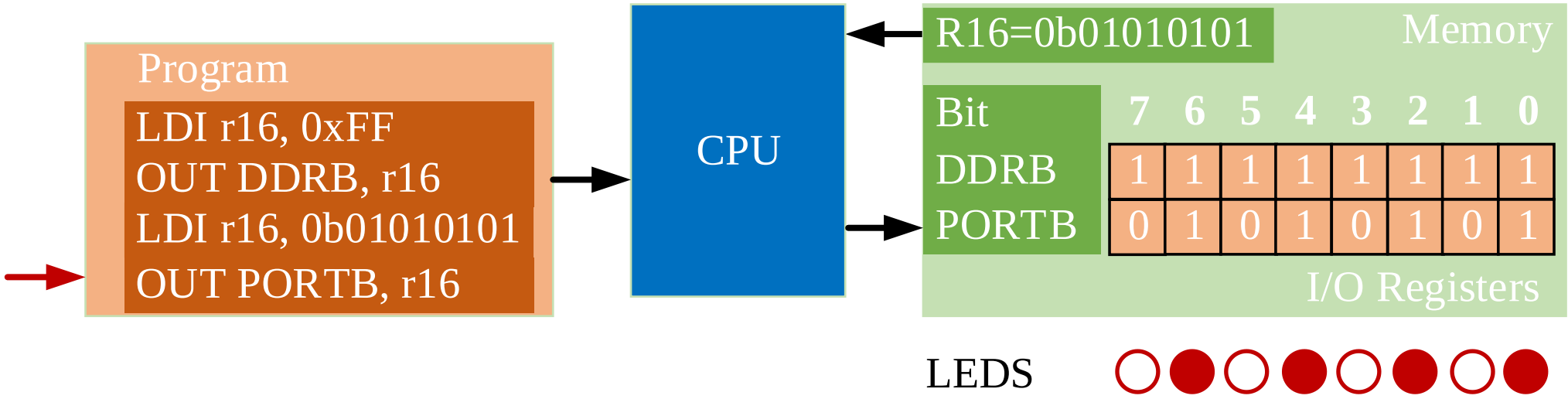
Write to a port - I

- Turn some pins on



Write to a port - II

- Turn some pins on

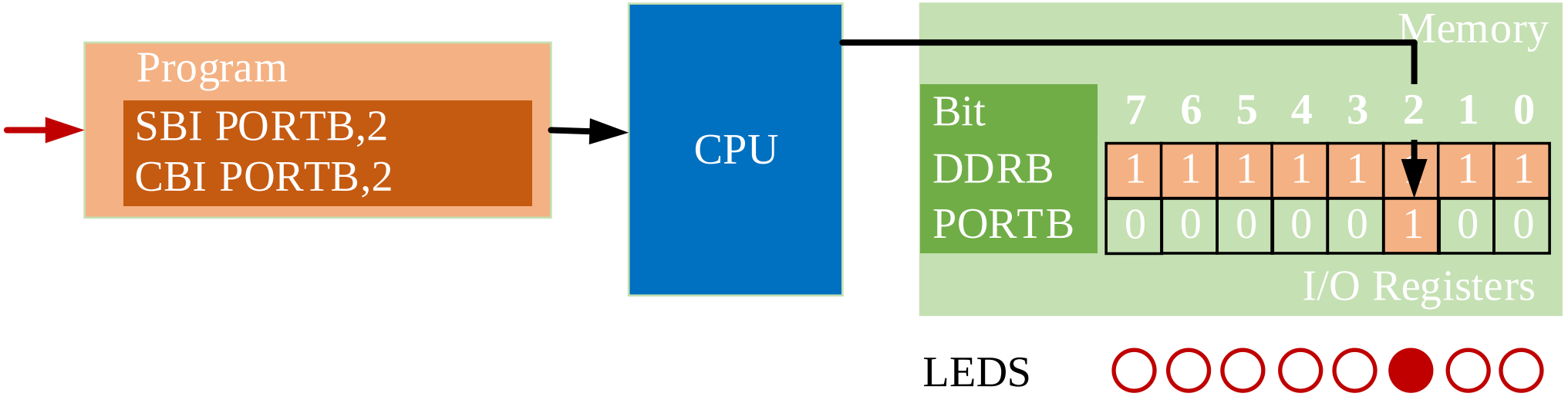


Single bit operations on ports

- Sometimes it’s convenient to manipulate individual bits of a port
- Control a single LED for instance
  - *SBI PortB, 3 //*Set bit 3 on PortB,
  - *CBI PortB, 3 //*Clear bit 3 on PortB

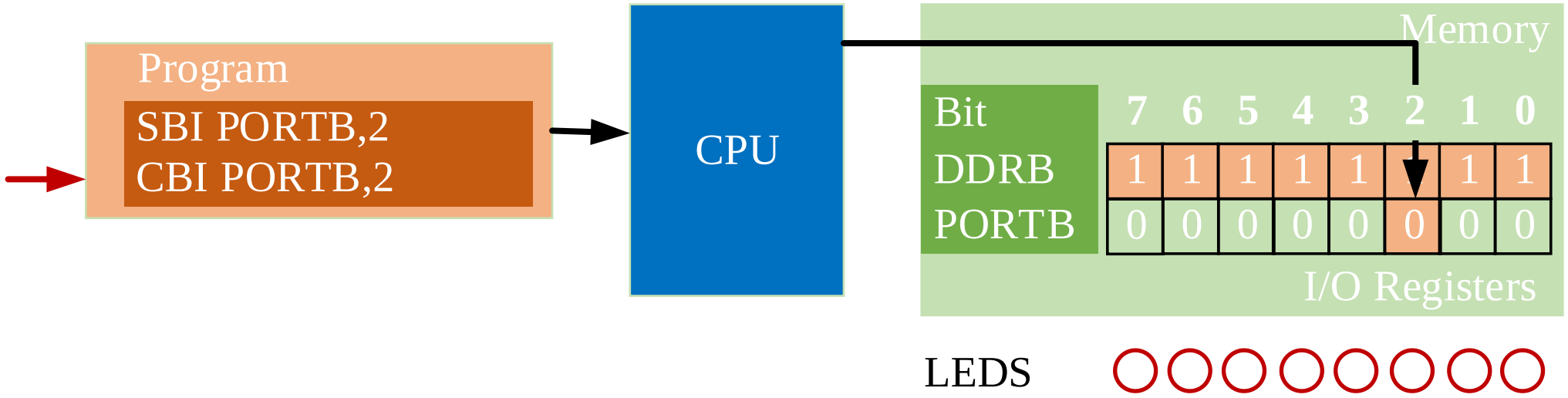
Single bit operation - I

- Assumed DDRB is set for output
- Set bit 2 (3rd bit) to on



Single bit operation - II

- Clear the 3rd bit to off



Toggling a port pin

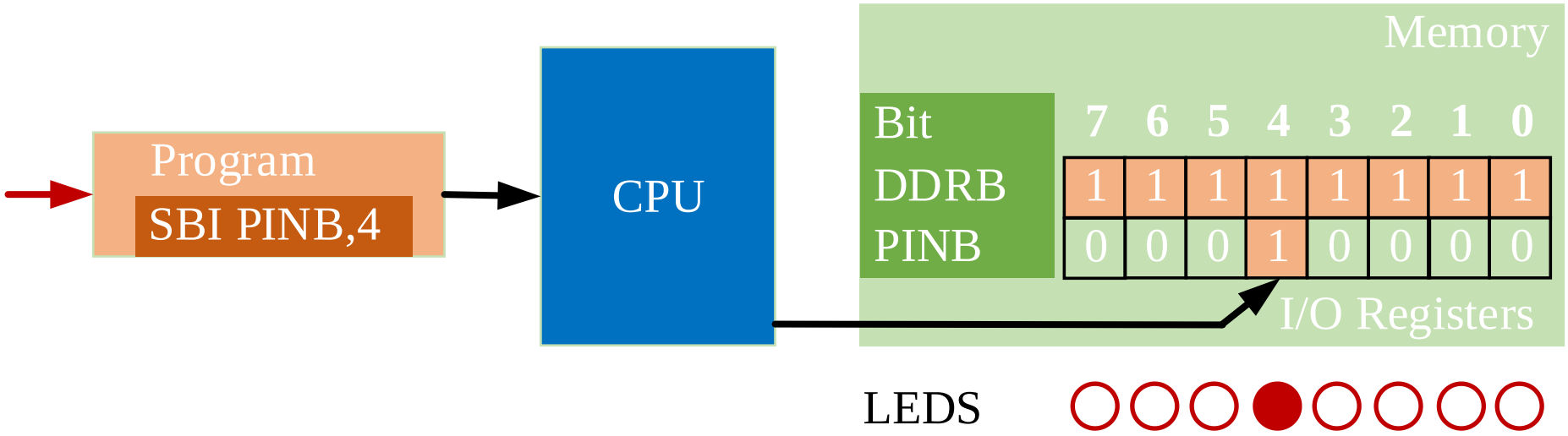
- Sometimes you don't know the state of a pin but you want it to change
  - For instance flashing an LED
- Write a 1 to the correct bit of the PINB register (for portB) toggles the state of that pin.

PINB-The Port B Input Pins Address

Bit	7	6	5	4	3	2	1	0	
0x03(ox23)	PINB7	PINB6	PINB5	PINB4	PINB3	PINB2	PINB1	PINB0	PINB
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

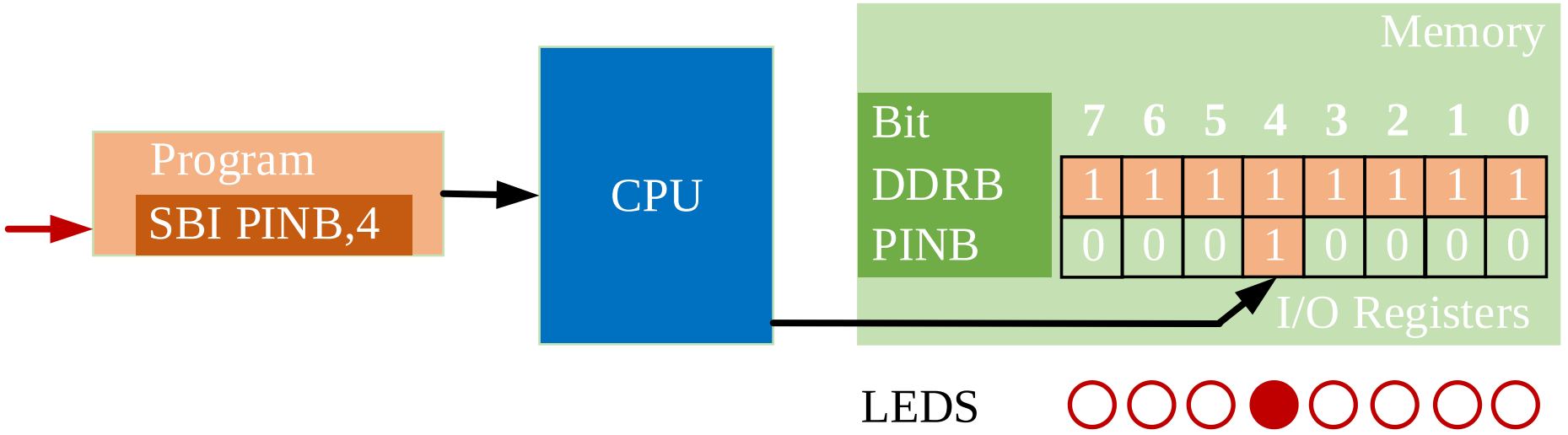
Toggling - I

- Setup DDRB in advance (to outputs)
- Write to pin 4 of PINB



Toggling - II

- Write to pin 4 of PINB

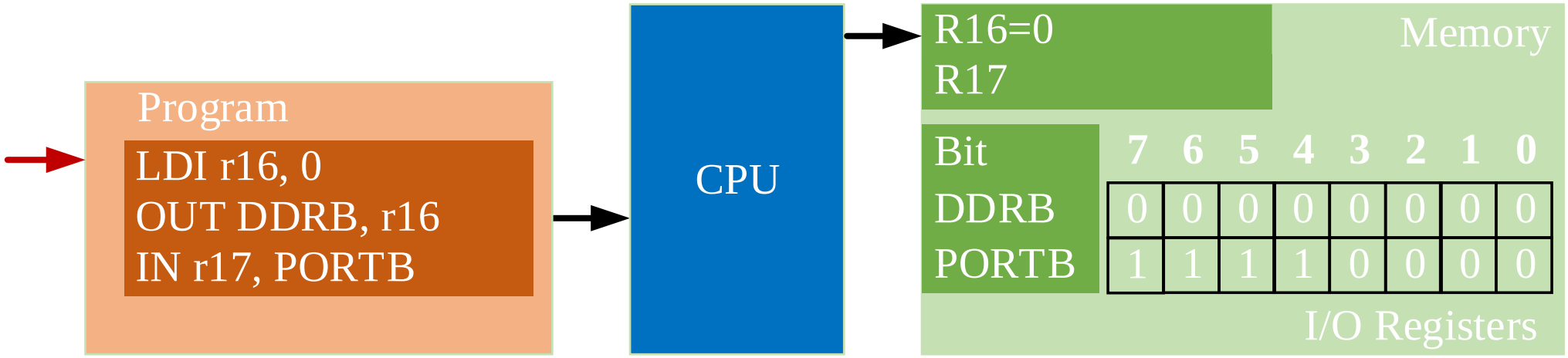


Reading from a port - I

- Get digital data INTO the chip
  - Switch for instance
- IN command copies a port state to a register
  - IN r16, PortB

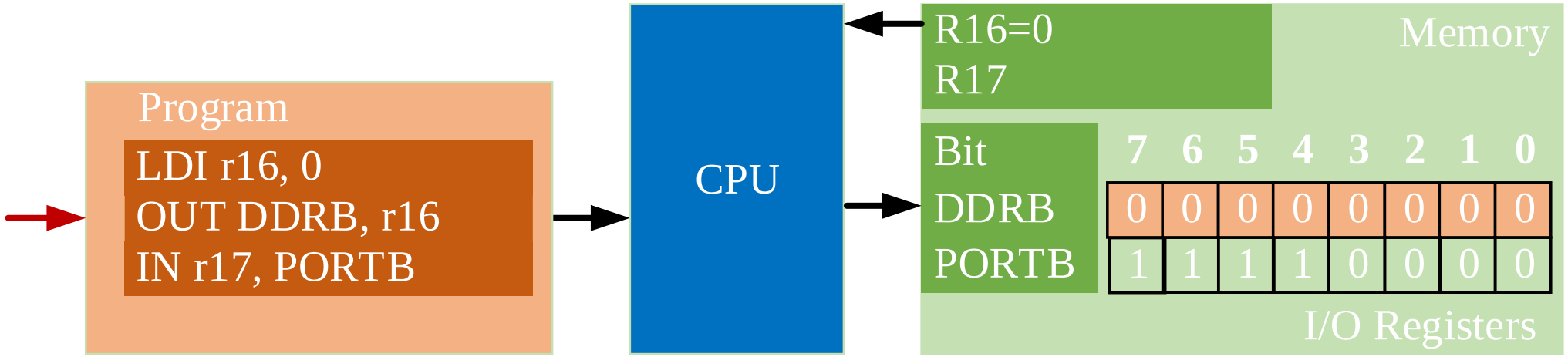
Reading from a port - II

- Write 0b00000000 or 0b0 or 0x00 or 0 to r16



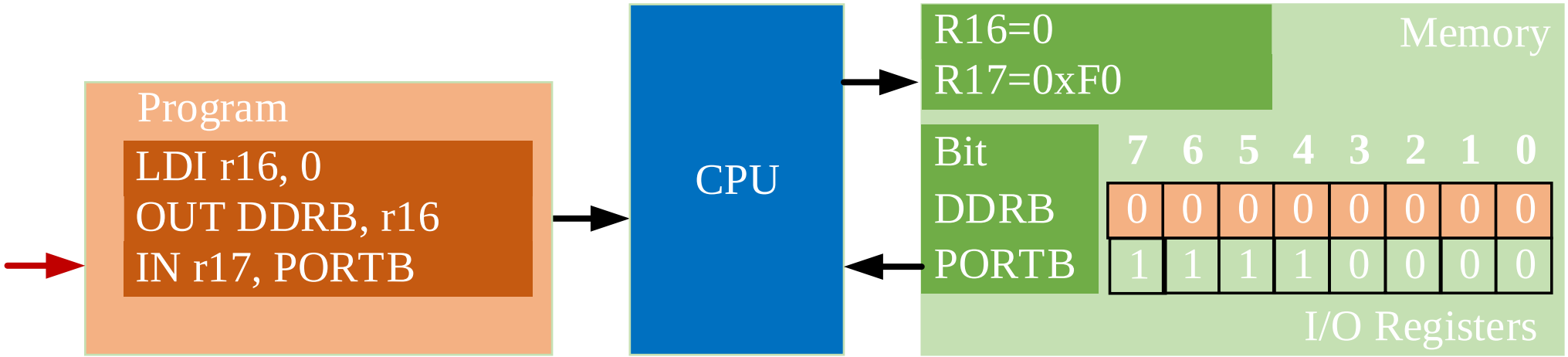
Reading from a port - III

- Set all pins as inputs



Reading from a port - IV

- Read the value of PortB to register r17



Next Lecture

- Overview of Arduino C